

AirSense Pakistan

Hardware Requirements and Prediction Framework

Technical Reference Document — Campus Pilot Phase 1

Field	Detail
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Scope	Phase 1: Rooftop Sensor Deployment and Prediction Framework

1. Introduction and Purpose

This document defines the complete hardware specification, sensor component roles, feature engineering framework, and recommended modelling approach for the AirSense Pakistan campus pilot deployment. It serves as the primary technical reference for the three-month rooftop sensor installation at Beaconhouse International College, Islamabad.

The document answers three core questions: What hardware is required and why? Is the proposed hardware sufficient for the forecasting objectives? And how does the hardware feed into the predictive modelling pipeline?

2. Hardware Sufficiency Assessment

2.1 Assessment Summary

Verdict: The proposed hardware stack is SUFFICIENT for Phase 1 campus pilot objectives.

The sixteen-component build covers all feature variables identified in the forecasting framework: three particulate fractions (PM1, PM2.5, PM10), three on-site meteorological variables (temperature, relative humidity, barometric pressure), a rain/wet-event flag, temporal features, and lag/rolling features derived in software. Wind speed and wind direction are supplied via free public weather APIs — a deliberate design choice that avoids sourcing low-cost local wind hardware that is not reliably available in Pakistani electronics markets.

2.2 Component-Level Sufficiency Check

The table below maps each forecasting feature to its hardware or data source and confirms inclusion status:

Feature / Variable	Status	Source and Notes
PM2.5 Concentration	INCLUDED	PMS7003 sensor. Primary target variable for all model training and inference.
PM10 Concentration	INCLUDED	PMS7003 sensor. Companion particulate predictor; used as lag feature input.
PM1 Concentration	INCLUDED	PMS7003 sensor. Additional fine particulate signal available from same device.
Temperature	INCLUDED	BME280 sensor. Atmospheric stability covariate.
Relative Humidity	INCLUDED	BME280 sensor. Affects particle growth, hygroscopic behaviour, and deposition.

Feature / Variable	Status	Source and Notes
Barometric Pressure	INCLUDED	BME280 sensor. Strongest identified predictor (Pearson $r = 0.342$ in baseline dataset).
Rain / Wet Event Flag	INCLUDED	Raindrop detection module. Binary flag for wet conditions; indicates particle washout events.
Wind Speed	API/HYBRID	Open-Meteo / WTTR.in public API. Field-validated monthly with handheld anemometer.
Wind Direction	API/HYBRID	Open-Meteo / WTTR.in public API. Source-region transport indicator.
Hour of Day	SOFTWARE	Derived from ESP32 system clock. Captures traffic and human activity diurnal patterns.
Day of Week	SOFTWARE	Derived from ESP32 system clock. Weekday vs. weekend emission patterns.
PM2.5 Lag Features	SOFTWARE	Computed in Python from sensor log. t-1 through t-24 lag values for autocorrelation modelling.
PM10 Lag Features	SOFTWARE	Computed in Python from sensor log. Companion lag predictor.
Rolling Means	SOFTWARE	Computed in Python. 3-hour and 6-hour rolling averages smooth noise and capture trends.
Peak Hour Flag	SOFTWARE	Derived from hour-of-day feature. Marks high-traffic periods (07:00-09:00, 17:00-20:00).

2.3 Known Limitations

- Wind speed and direction are not measured in real time on-site. The hybrid strategy (public API + monthly field validation) mitigates this but does not fully replace local anemometry. This is an accepted limitation for Phase 1.
- The PMS7003 sensor accuracy can degrade under very high humidity (above 90% RH) due to particle hygroscopic growth. Data quality control procedures must flag and handle high-humidity readings appropriately.
- Rooftop temperatures in Pakistani summers can approach the operational limit of the sensor enclosure. The white radiation shield is essential and must be installed correctly.
- Local source events such as generator operation, construction, or road congestion adjacent to the campus should be logged manually as contextual notes to help interpret anomalous readings in the data.

3. Component Specifications

3.1 Sensing Layer

PMS7003 Particulate Sensor

Parameter	Detail
Measurement Output	PM1.0, PM2.5, PM10 concentration in micrograms per cubic metre
Measurement Principle	Laser light scattering (optical particle counter)
Measurement Range	0 to 1,000 micrograms per cubic metre (PM2.5)
Accuracy	+/- 10 micrograms per cubic metre for readings up to 100; +/- 10% above 100
Operating Temperature	minus 10 to plus 60 degrees Celsius
Interface	UART serial (3.3V / 5V compatible)
Warm-Up Time	30 seconds to stable reading
Data Rate	Up to 1 reading per second; 1-minute polling recommended for this deployment
Why Selected	Industry-standard low-cost PM sensor; wide deployment in DIY air quality research; proven reliability in hot and dusty environments

BME280 Environmental Sensor

Parameter	Detail
Measurement Output	Temperature (degrees Celsius), Relative Humidity (%), Barometric Pressure (hPa)
Temperature Range	minus 40 to plus 85 degrees Celsius; accuracy +/- 0.5 degrees Celsius
Humidity Range	0 to 100% RH; accuracy +/- 3% RH
Pressure Range	300 to 1,100 hPa; accuracy +/- 1 hPa
Interface	I2C or SPI (I2C used in this design)
Sampling Rate	Up to 2 Hz; 1-minute polling adequate
Why Selected	Three critical forecasting covariates in one compact, low-power module; widely supported in Arduino and ESP32 ecosystems; barometric pressure identified as strongest predictor in the AirSense baseline model

Raindrop Detection Sensor Module

Parameter	Detail
Output Type	Digital HIGH/LOW (wet / dry) + analog voltage level
Role in Framework	Provides a binary rain-event flag for each timestamp; wet conditions typically precede PM2.5 washout events
Operating Voltage	3.3 V to 5 V
Why Selected	Simple, cheap, and effective proxy for precipitation events; captures the cleansing effect of rain on particulate concentrations without requiring a calibrated rain gauge

3.2 Control and Communication Layer

ESP32 WROOM-32D Microcontroller

Parameter	Detail
Processor	Dual-core Xtensa LX6 at 240 MHz
WiFi	802.11 b/g/n (2.4 GHz) — built-in; no separate module required
Bluetooth	Bluetooth 4.2 — available but not used in this design
GPIO Pins	34 programmable GPIO; supports UART, I2C, SPI, ADC
Operating Voltage	3.3 V logic; 5 V input via USB / power supply
Role in System	Polls all sensors every 60 seconds; writes to microSD; pushes JSON data packet to cloud endpoint via WiFi; maintains RTC-based timestamps
WiFi Strategy	Connects to campus WiFi network; campus extenders extend coverage to rooftop installation point
Why Selected	Built-in WiFi eliminates the need for a separate network module; well-documented; large community support; lowest-cost solution for this combination of capabilities

3.3 Data Storage and Power

Component	Specification and Role
MicroSD + Reader	Arduino-compatible SPI microSD module with Samsung EVO Plus 16 GB card. Writes one CSV row per polling interval. Provides local data backup independent of WiFi availability. Estimated storage capacity for 1-minute intervals: approximately 18 months.
Power Supply	5V 2A mains power supply. Provides continuous, stable power. Requires mains socket access on or near rooftop. The 10m extension cable addition accommodates distances up to 10 metres from nearest socket.

Component	Specification and Role
Battery Backup	Not included in Phase 1 budget. If campus has UPS coverage for rooftop circuits (or generator backup), data continuity is adequate. If not, the microSD card preserves all data locally during outages for later upload.

3.4 Physical Protection

Component	Specification and Role
Weather Enclosure	IP65-rated waterproof DIY enclosure. Houses ESP32, BME280 module, SD reader, and power circuitry. The PMS7003 sensor inlet is positioned to allow air flow while the main electronics are fully sealed.
Radiation Shield	White plastic plate set forming a DIY Stevenson screen over the BME280 sensor. Prevents solar radiation from directly heating the sensor housing, which would bias temperature and humidity readings upward. Essential for rooftop deployment.
Silicone Sealant	Applied to all cable entry points on the enclosure. Prevents moisture ingress through wiring gaps, which is a common failure mode in outdoor low-cost sensor deployments.
Mounting System	Stainless hose clamps and M4/M6 stainless fasteners for attachment to rooftop parapet or railing. Stainless steel specified for corrosion resistance in exposed outdoor conditions.

4. Prediction Framework

4.1 Target Variable and Forecast Horizons

Target Variable: PM2.5 concentration (micrograms per cubic metre)
Forecast Horizons: 1-hour ahead, 3-hour ahead, 6-hour ahead, 24-hour ahead

4.2 Full Feature Set

Feature	Source	Role in Model
PM2.5 Concentration	PMS7003	Primary target; also used as lag feature input
PM2.5 Lag (t-1 to t-24)	Derived from sensor log	Captures persistence and autocorrelation — most important predictor class
PM10 Concentration	PMS7003	Companion particulate; strong correlated predictor
PM10 Lag (t-1 to t-24)	Derived from sensor log	Extends particulate lag feature space

Feature	Source	Role in Model
PM1 Concentration	PMS7003	Available from same sensor; useful for ultra-fine particle dynamics
Temperature	BME280	Atmospheric stability and boundary layer height covariate
Relative Humidity	BME280	Particle growth factor; elevated RH can artificially inflate PM readings
Barometric Pressure	BME280	Strongest single predictor ($r = 0.342$); reflects weather system state and dispersion potential
Rain Event Flag	Raindrop Sensor	Binary covariate; rain events typically produce sharp PM2.5 decreases
Wind Speed	Public API (Open-Meteo)	Dispersion and transport covariate; higher wind typically reduces local PM
Wind Direction	Public API (Open-Meteo)	Source-region indicator; identifies when readings are influenced by industrial or traffic upwind sources
Hour of Day	Software (ESP32 RTC)	Captures diurnal patterns in traffic, cooking, and industrial activity
Day of Week	Software (ESP32 RTC)	Weekday vs. weekend emission pattern differentiation
Rolling Mean 3-hr	Derived from sensor log	Smooths short-term noise; represents recent pollution trend
Rolling Mean 6-hr	Derived from sensor log	Captures medium-term trend; useful for 6-hour and 24-hour horizon models
Peak Hour Flag	Derived from hour feature	Binary flag for morning and evening traffic peaks
Weekend Flag	Derived from day-of-week	Binary; different emission profiles on Fridays and weekends in Pakistan

4.3 Data Collection and Synchronisation

- The ESP32 polls all on-site sensors every 60 seconds and writes a timestamped record (Unix epoch + ISO 8601) to the microSD card and to the cloud endpoint.
- Public weather API data (wind speed, wind direction, rainfall cross-check) is fetched hourly and merged with on-site data by timestamp.
- The primary working dataset for model training is aggregated to one-hourly resolution by averaging all 1-minute readings within each hour.
- Data quality control is performed before any model training: impossible values are removed, short gaps (up to 2 hours) are filled by linear interpolation, long gaps (greater than 6 hours) are flagged and excluded from training.

4.4 Recommended Modelling Approach

Step 1 — Baselines

Begin with a persistence baseline (forecast equals the last observed value) and a linear regression model. These establish the minimum performance bar that all subsequent models must beat.

Step 2 — Tree-Based Ensemble Models

Progress to Random Forest, XGBoost, and LightGBM. These models handle the non-linear interactions between meteorological features and particulate concentrations effectively and are robust to moderate dataset sizes. They are the recommended primary models for the campus pilot.

Step 3 — Sequence Models (Optional, Later Phase)

LSTM and GRU sequence models can be explored once the dataset exceeds 5,000 clean hourly observations and the team has validated the quality of the feature engineering pipeline. They are not required for a strong pilot-phase result and carry higher implementation complexity.

Step 4 — Validation Strategy

Critical: Use walk-forward (rolling-origin) validation, not random train-test split.
Air quality data is time-ordered. A random split would allow models to train on future data and evaluate on past data, producing falsely optimistic results. Walk-forward validation mirrors real operational forecasting: train on all data up to time T, evaluate on time T+1 to T+H, then advance the window forward. This is the only valid method for evaluating operational forecast performance.

Step 5 — Evaluation Metrics

Metric	Definition and Usage
RMSE	Root Mean Square Error — primary metric; penalises large errors
MAE	Mean Absolute Error — interpretable in same units as PM2.5 (micrograms per cubic metre); current baseline: 49.27
R2	Coefficient of determination — current baseline: 0.412; target: above 0.60 on campus dataset
MAPE	Mean Absolute Percentage Error — use cautiously; unstable near zero concentrations
Classification Metrics	When PM2.5 thresholds are relevant (e.g. WHO levels), evaluate precision and recall for threshold exceedance detection

Step 6 — Interpretability

Apply SHAP (SHapley Additive exPlanations) values to identify which features drive model predictions for each forecast horizon. Interpretability is important for the operational decision layer — clients and regulators will need to understand why the system issued a particular alert.

5. Operational Decision Layer

The operational decision layer is the interface between the predictive engine and the end user. It maps forecast PM2.5 bands to specific, actionable business guidance. During the campus pilot, this layer runs in evaluation mode — the team assesses the quality and practicality of each alert generated rather than serving them to external clients.

Trigger / Condition	Alert Category	Action in Pilot Mode
PM2.5 Forecast Band	Alert Level	Pilot Action
Below 35 microg/m3	Good / Acceptable	No action required. Log as normal operating condition.
35 to 75 microg/m3	Moderate	Issue advisory. In pilot mode: log advisory trigger and note whether the team would have acted.
75 to 150 microg/m3	Unhealthy	Issue alert. In pilot mode: log alert trigger, evaluate specificity of the forecast.
Above 150 microg/m3	Hazardous	Issue urgent alert. In pilot mode: log and compare with actual observed PM2.5 on same day.
High PM + Low Wind	Elevated Risk	Secondary alert combining PM forecast with low wind speed from API feed. Log and evaluate combined trigger accuracy.
Rain Event Forecast	Cleansing Event	Forecast PM reduction expected. Log prediction vs. observed post-rain PM2.5 decline.

At the end of the three-month pilot, the team will analyse all generated alerts, categorise them as true positives, false positives, or missed events, and use the results to refine the threshold logic before commercial deployment.

6. Deployment Checklist

6.1 Pre-Deployment

Task	Completion Criteria
Hardware Assembly	Assemble sensor circuit on breadboard. Test all sensor readings via serial monitor. Verify data types and value ranges.
Code Upload	Upload polling and WiFi push firmware to ESP32. Test with local WiFi. Verify JSON packet structure.
Cloud Endpoint	Deploy data receiver API on free-tier cloud host. Test end-to-end data flow from sensor to database.

Task	Completion Criteria
SD Card Test	Confirm CSV write to microSD. Confirm file readable on PC. Verify timestamp accuracy.
Enclosure Assembly	Mount all electronics in enclosure. Route cables. Apply silicone sealant to all entry points. Fit radiation shield over BME280.
Site Survey	Visit rooftop. Identify optimal mounting location. Confirm power socket proximity. Test WiFi signal strength at location.

6.2 Installation Day

Task	Completion Criteria
Mount Sensor Unit	Attach enclosure to parapet or railing using stainless clamps and fasteners. Ensure unit is vertical and radiation shield faces away from direct morning sun.
Connect Power	Run extension cable from rooftop socket to unit. Confirm LED indicators show ESP32 is booted and WiFi-connected.
Verify Data Flow	Check cloud database for incoming records within 5 minutes of power-on. Confirm timestamp, PM2.5, temperature, humidity, and pressure fields are populated.
Baseline Reading	Record first 30 minutes of readings manually for comparison. Note time of day, weather conditions, and any nearby activity.
Document Installation	Photograph the final installation for the project record and end-of-pilot report.

6.3 Monthly Maintenance

Task	Procedure
Sensor Cleaning	Gently clean PMS7003 inlet of dust accumulation using compressed air or soft brush. Do not disassemble sensor.
SD Card Check	Extract SD card. Verify all expected records are present. Copy data to secure backup. Reinsert card.
Enclosure Integrity	Check silicone seals for cracking or peeling. Inspect clamp tightness. Look for signs of moisture ingress or heat distortion.
Wind Validation	Use handheld anemometer to take 3 spot readings over 10 minutes. Compare with API wind speed data for the same period. Log the comparison result.
Data Quality Review	Run QC script on the past month's data. Report gap rate, outlier count, and any sensor drift indicators.

Appendix A — Pin Wiring Reference

Sensor / Module	ESP32 Pin	Function
PMS7003 TX	ESP32 GPIO16 (UART2 RX)	Serial data from sensor to controller
PMS7003 VCC	ESP32 VIN (5V)	PMS7003 requires 5V; use VIN pin on ESP32 board
PMS7003 GND	ESP32 GND	Common ground
BME280 SDA	ESP32 GPIO21 (I2C SDA)	I2C data line
BME280 SCL	ESP32 GPIO22 (I2C SCL)	I2C clock line
BME280 VCC	ESP32 3.3V	BME280 operates at 3.3V
BME280 GND	ESP32 GND	Common ground
SD Reader MOSI	ESP32 GPIO23 (SPI MOSI)	SPI data out
SD Reader MISO	ESP32 GPIO19 (SPI MISO)	SPI data in
SD Reader SCK	ESP32 GPIO18 (SPI SCK)	SPI clock
SD Reader CS	ESP32 GPIO5	SPI chip select
Rain Sensor DO	ESP32 GPIO34 (ADC input)	Digital or analog rain signal
Rain Sensor VCC	ESP32 3.3V	Power
Rain Sensor GND	ESP32 GND	Common ground